From Published Evidence to Analyzable Biomedical Data with a Scientific Agent

Working research draft — 4 November 2026

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Abstract

Biomedical studies often lack measurements needed for a new analysis, while relevant distributions and associations already appear in the literature. We study whether a scientific Agent can turn this partial evidence into useful virtual data without individual-level construction records. The Agent interprets findings under their study populations and analysis definitions, chooses a joint generating law, translates usable statistics into numerical constraints and records assumptions for missing relationships. Statistical tools execute the model and return diagnostic feedback. In a developmental study with CHNS and ENSANUT references, 20 Agent and four statistical constructions produced 72 complete tables. Additional relational evidence reduced mean absolute discrepancy for five of six specified coefficients. Separate summary-assisted and published-model experiments supported closer joint-statistic reconstruction and longitudinal analysis of 11,509 virtual subjects. An open-search case connected independently retrieved evidence to three complete 3,000-person tables, with useful agreement for several requested relationships and remaining conditional discrepancies. A pre-generation sensitivity analysis identified an unreported dependence that could reverse one conditional coefficient while leaving other relationships unchanged. These results establish concrete scientific uses of evidence-guided virtual data and show how Agent-authored models can expose the missing evidence that limits an intended analysis.

1 Introduction

A research question can be well defined before the necessary biomedical records are available. A cohort may never have measured a biomarker, another study's individual records may be inaccessible, or no accessible dataset may contain the required combination of variables. Published means, distributions, adjusted associations and subgroup findings nevertheless contain relevant information. A researcher could use a virtual dataset assembled from that evidence to run an intended analysis, inspect its implications and plan subsequent data collection. The practical output is a table that ordinary scientific software can analyze, accompanied by the assumptions that make the table possible.

The difficult step lies between a published finding and a joint data-generating process. An age-adjusted regression coefficient, a marginal correlation and a grouped contrast describe different statistical quantities. Their populations, measurements and adjustment sets also differ. Even two correctly extracted coefficients cannot generally be inserted unchanged into one joint model. Further relationships needed to generate a complete table may never have been reported. An automated system therefore has to make scientific modeling decisions, in addition to extracting numbers or sampling rows.

We investigate an evidence-guided Agent workflow for this task. Given a population description, requested variables and scientific analyses, the Agent acquires or receives relevant studies, chooses the numerical role of each finding, authors a joint model and inspects the resulting data. Deterministic tools perform calculations and sampling. A request may lead to a different generator when the target population, available evidence or intended analysis changes. The scientific question is whether the resulting data preserve useful properties for the requested analysis, and which properties still depend on unsupported completion or transport assumptions.

This focus builds on established synthesis methods. Filshtein et al. generate complete cohorts using source-specific models estimated by collaborating analysts [1]. SynthIPD constructs survival records and categorical covariates from published trial material [2]. Multilevel network meta-regression relates aggregate findings to individual-level models under population assumptions [3]. LLM-based systems additionally automate consequential construction decisions: G-Sim combines simulator authorship, numerical calibration and structural refinement [4]; LLMSynthor plans micro-record generation from aggregate information and uses discrepancy feedback [5]; SynthCraft supports synthetic-data workflows in healthcare [6]. SPADA separates an LLM-induced dependency graph from statistical sampling fitted to an observed table [15]. Our investigation concerns the scientific work required to connect a biomedical analysis request, study-qualified partial evidence and an analyzable joint dataset when individual construction records are absent.

The paper makes three contributions. **First**, it implements an Agent workflow in which the scientific author selects and numerically uses evidence, chooses the generating law and makes a documented retain-or-revise decision. **Second**, it makes evidence-to-model decisions inspectable: we demonstrate the translation of separate adjusted projections into joint coefficients and the identification of a missing dependence that controls a requested conditional effect. **Third**, it evaluates the resulting data through ordinary scientific analyses

under external-evidence, supplied-summary and published-model information settings, together with a completed open-search construction.

The experiments provide complementary evidence rather than one pooled benchmark. The larger study uses curated literature to evaluate an information contrast. Other settings ask whether accessible summaries can become useful joint data, whether a published model can support a longitudinal analysis, and whether an actual search can lead to a complete data product. The outcome is a practical evidence-to-data procedure with demonstrated analysis-specific uses and explicit boundaries. Ordinary Agents are valid implementations; specialized training and multi-Agent collaboration are not prerequisites.

2 Problem: evidence-conditioned data for a scientific request

Let a request contain the target population, eligibility and measurement definitions, desired variables, permitted observations and intended analyses. A study evidence record identifies its population, measurement scale, statistical operator and adjustment set, reported estimate, available uncertainty and provenance. A set of such records provides only partial information about the joint law needed by the request.

The Agent's construction is an executable generator and its scientific specification:

```
(generator, evidence uses, completion assumptions)
     = Agent(request, evidence, optional
    observations)

virtual data = generator(number of rows, fixed
    random seed)

analysis results = requested scientific analyses(
    virtual data).
```

This notation describes the implemented procedure; it does not introduce a trained conditional model or a new optimization objective. In complete-cohort generation, the optional observations are empty and the generator supplies every requested variable. In enrichment, observed covariates remain fixed while an unmeasured variable is generated. The present principal experiments concern complete cohorts. Earlier enrichment experiments are supporting evidence under their own versions; missing-cell completion remains an unevaluated mode.

We distinguish three legitimate uses of evidence. A reported statistic can constrain the model numerically, provide a check on its implications, or remain contextual when its meaning or applicability is insufficiently resolved. Algebraic consequences of accepted statistics are derived evidence, whereas unspecified dependence and source-to-target transport remain assumptions. A numerical constraint from another population does not automatically identify the corresponding target coefficient.

Quality is assessed at the level of scientific analysis. For an analysis statistic, such as an adjusted regression coefficient, we report the generated estimate alongside the relevant sample reference, effect direction, magnitude and available uncertainty. Source-summary agreement measures reconstruction of supplied information. Agreement with a target result not supplied to the author addresses a different question. Neither requires individual-row recovery or exact agreement with one real sample. A complete virtual table can support more than one analysis, but different uses may have different evidential support.

This problem is generally underidentified. Several joint laws can reproduce available margins and associations while implying different conditional or subgroup effects. The system consequently delivers a data product together with its construction assumptions. More generated rows improve Monte Carlo precision under a law; they do not establish unreported biological relationships. Identifying which missing relationship matters for a requested use is itself a useful outcome of constructing an explicit model.

3 Method

3.1 Qualify evidence against the request

The workflow begins with a structured task containing population, era, measurements, units, eligibility and visible scientific questions. It accepts curated evidence and Agent-directed acquisition. When retrieval is enabled, the Agent can formulate queries and select primary studies relevant to its evidence gaps. The retained record distinguishes original full text, partial access and bibliographic metadata, and records study overlap or exclusions.

Each extracted finding is interpreted through its original analysis definition. Logistic disease coefficients cannot silently become continuous-biomarker slopes; grouped contrasts cannot become per-unit effects; Pearson and Spearman associations require different numerical treatment. Likewise, a change of measurement scale supplies an algebraic conversion, not additional information about transport. Source comparisons are classified as directly supported calculations, explicitly assumption-qualified approximations or unavailable operations. Reconstructing every original study population is not a prerequisite for generating a target dataset.

In the principal study, researchers supplied audited source packets and the Agent selected their scientific use. In the open-search case, one Agent acquired a five-study packet without bibliographic seeds, and a fresh constructor used that reviewed packet. These are observed implementations of the shared procedure, not claims that every historical experiment used autonomous retrieval. The host dispatched Agents and a coordinator managed tool access, execution and closure.

3.2 Author a joint law and explain its numerical use

The scientific Agent chooses a probability family, dependencies, numerical parameters and any unreported completion assumptions. It supplies generator code, a model specification,

evidence-use records and a decision explanation. A cited relationship counts as numerical use only when the implemented law or an explicit calculation shows how it enters a constraint or check. The coordinator does not insert target-derived coefficients or desired signs into the scientific model.

A worked construction illustrates the distinction. Jansen et al. report separate age/sex-adjusted HbA1c projections for BMI and glucose [9]. For residual BMI and glucose variables with SDs `sB` and `sG` and assumed correlation `r`, a joint response equation with coefficients `thetaB` and `thetaG` implies

```
cB = thetaB + thetaG * r * sG/sB

cG = thetaG + thetaB * r * sB/sG.
```

CHNS E1 constructor A01, selected by ordinal identifier rather than performance, solved these equations using reported slopes `cB = 0.13` and `cG = 2.29`, residual SDs 4.07611 and 0.49694, and the declared assumption `r = 0.25`. The resulting coefficients were 0.0642173 and 2.1583154 in IFCC response units. Multiplying by 0.09148 gave 0.00587460 and 0.19744269 in HbA1c percentage-point units [11]. Thus the Agent used both projections without assigning them unchanged to one joint equation. Residual correlation, factor distributions, age/sex contributions and target transport remained assumptions.

The worked trace links this derivation to frozen code, an unfitted source-correlation check and the constructor's target BMI mismatch. The mathematical identities are established statistics. The implemented contribution is the Agent's evidence-specific selection, derivation and use of them within data construction, with the consequences made inspectable.

3.3 Execute, inspect and finalize

Deterministic tools execute the authored program through an isolated WSL/bubblewrap adapter, calculate the allowed statistics and return source/generated-data feedback. Semantic checks assess the declared measurement, operator and role. Separate reviews inspect numerical calculations, support and required finite moments. These checks establish implementation properties; source extraction, biological plausibility and population transfer require their own evidence.

The Agent can revise or retain its model within the task budget. A valid no-op ends the construction; the process does not require an extra revision to manufacture improvement. Each candidate, final seed allocation and evaluation contract is fixed before the corresponding execution or comparison. The final data and model assumptions are delivered together, so the intended analyses can be run in standard software. Author access boundaries are procedural even though generator execution is isolated.

The shared scientific procedure has separately frozen experimental implementations, source packets, feedback opportunities and evaluators. Main-study requests generated complete age, sex, BMI, glucose and HbA1c tables. The longitudinal case used a separate native mixed-model evaluator. The latest controller bound one specific complete-cohort request; general standalone interpretation and execution of arbitrary natural-language requests are not established by these experiments.

4 Experiments

We organize evaluation around four questions: **Q1**, does additional relational evidence support useful target analyses? **Q2**, can accessible summaries become a coherent joint dataset? **Q3**, can the procedure construct longitudinal data for a published analysis? **Q4**, can open evidence acquisition lead to an analyzable cohort and reveal an analysis-sensitive evidence gap? Table 1 separates their information conditions. Individual-level records were not supplied to any constructor in these settings.

4.1 Evaluation design

CHNS included 6,631 selected adults aged 35–74 with complete age, recorded sex, BMI, central glucose and HbA1c. Analyses were unweighted selected-sample estimates; diabetes/treatment exclusions were not imposed and individual fasting adherence was unverified. ENSANUT included 6,931 selected adults aged 25–79 under fixed fasting, prior-diagnosis, glucose and anthropometry criteria. It assigned equal mass to eight sex-by-age cells rather than estimating national survey effects. These populations, measurement units and estimators remain distinct.

Q1 allocated five fresh Agent contexts to each case/condition and one fixed statistical specification to each case/condition. Each construction had three 3,000-row final attempts, seeds 5101–5103, and at most two exploratory row-producing attempts. The fixed statistical specifications used the same admitted numerical information and explicit completion assumptions. One prescribed specification versus five independently authored Agent laws is an asymmetric descriptive comparison. Matching author labels across E0/E1 do not imply pairing.

All constructions closed before common reference evaluation. Each case retained three focal regressions and 27 CHNS or 28 ENSANUT marginal/joint diagnostics. Four focal analyses had source-analogous projections; CHNS conditional age and ENSANUT conditional BMI were neither directly supplied as target coefficients nor identified by the admitted packet. For each focal statistic, absolute reference discrepancy was averaged over three seeds within each constructor, then across five Agent constructors. E1–E0 compared these arm means. Three-seed ranges quantify sampling under a fixed law. CHNS community-cluster CR1-t intervals and ENSANUT prespecified cluster-resampling stability bands contextualize their sample estimates; neither defines equivalence.

Both target families had been exposed during earlier development. Fresh author contexts restrict construction inputs but do not restore untouched clinical validation. Separately written numerical audits checked generated statistics and frozen reference mappings. Complete eligibility, formulas, reference intervals, allocations, failures and diagnostics are preserved in the extended methods and results. The principal comparison,

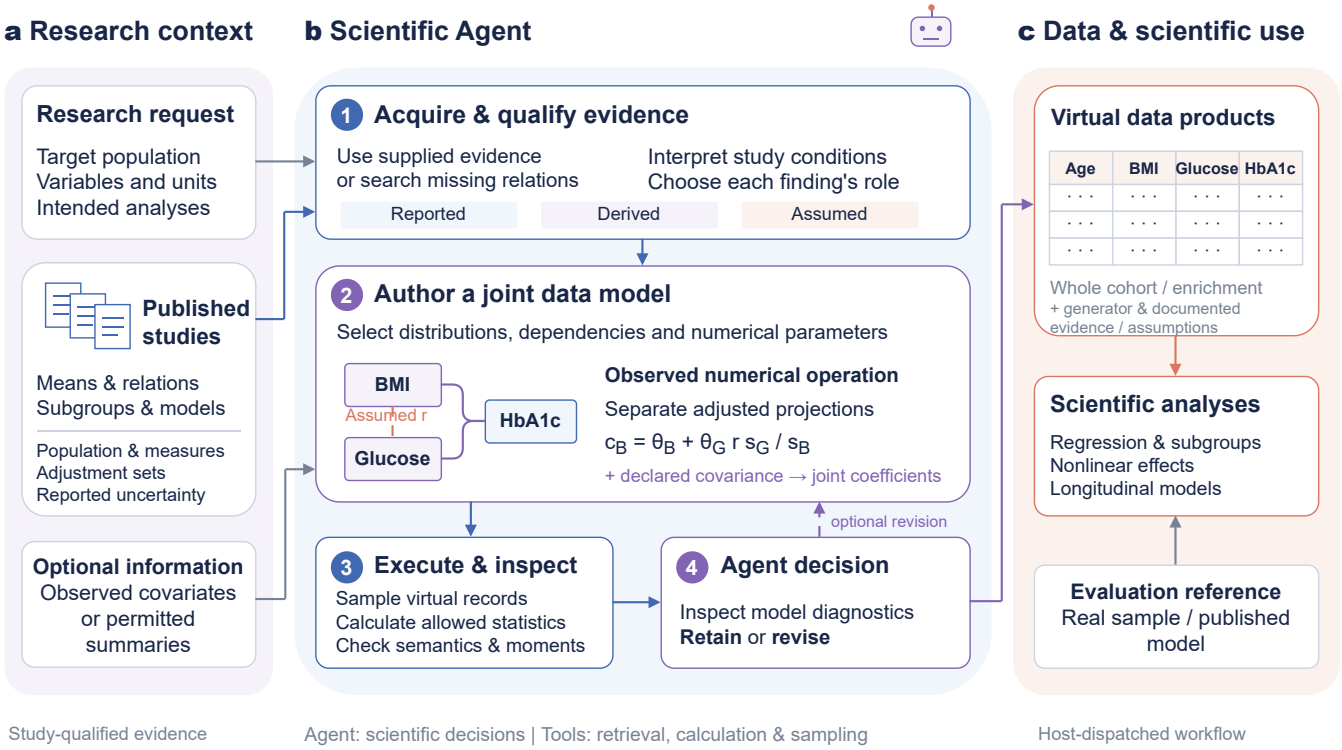


Figure 1: Evidence-guided Agent construction and scientific use. The three regions connect research context, scientific construction and a reusable data product. Curated or retrieved findings retain study conditions and reported, derived or assumed status; their use roles are constraint, check or context. The Agent selects a joint law and numerical parameters. The inset shows one of the two projection identities used in the A01 derivation under an assumed residual covariance; its lines depict model dependence, not biological causation. Tools sample records and calculate permitted diagnostics, followed by an optional retain-or-revise decision. Whole-cohort generation is primary and the principal constructors receive no individual records; earlier experiments support enrichment. Permitted observations or summaries enter only in the corresponding setting. The table uses placeholders. Reference data support evaluation under each experiment's information conditions. Host dispatch and optional revision do not imply standalone autonomy or demonstrated revision gain.

Table 1: Evaluation questions and information conditions. A construction is an authored model; seeds sample that model. Reused CHNS analyses are not additional target families.

Question and setting	Construction inputs	Constructions and final data	Reference role
Q1: external evidence, CHNS and ENSANUT	Population/measurement requests; E0 basic/marginal evidence or E1 additional relational evidence	20 Agent + 4 statistical constructions; 72 tables of 3,000 rows	Previously exposed selected-sample references; target coefficients not supplied to fresh authors
Q2: CHNS summary assistance	Literature in L; literature plus 30 target summaries and metadata in R	6 Agent constructions; 18 distinct final tables of 3,000 rows	Reconstruction of explicitly supplied information in R
Q3: HRS published model	Published model artifacts and requested longitudinal analysis	1 Agent + 1 reconstructed published law; 6 valid tables, each 11,509 subjects at 4 visits	Published-model reconstruction; no HRS participant records
Q4: open search and construction	Five Agent-retrieved, reviewed studies and a population/analysis request	1 fresh constructor; 3 final tables of 3,000 rows	Reused CHNS aggregates after closure; one developmental workflow case

summary study, longitudinal example and later workflow units are not pooled into a single success score.

4.2 Q1: external relations improve several requested analyses

All 24 Q1 constructions produced their three scheduled final tables. The 72 final tables were distinct and supported every requested regression and diagnostic; seven exploratory tables brought physical executions to 79. Additional relational evidence reduced constructor-averaged mean absolute discrepancy for five of the six designated coefficients (Table 2). CHNS glucose discrepancy decreased from 0.134019 to 0.024079 HbA1c percentage points per mmol/L. All three ENSANUT contrasts improved. CHNS BMI discrepancy increased from 0.020380 to 0.031557.

The effect displays explain what these changes mean. CHNS E1 glucose constructor means were 0.363363–0.414189 around the 0.402093 reference. Its BMI means were positive but lower, 0.012400–0.045757 versus 0.054946. Conditional-age means ranged from −0.000509 to 0.007393 versus 0.008107; one author retained a neg-

Table 2: Complete main-study information contrast. Discrepancies retain the units of each row; they are not pooled. The reference response is HbA1c percentage points. E1–E0 uses unrounded quantities before display rounding.

Case / focal analysis	Predictor increment	E0 mean absolute discrepancy	E1 mean absolute discrepancy	E1 − E0
CHNS BMI, adjusted age/sex	1 kg/m²	0.020380	0.031557	+0.011178
CHNS glucose, adjusted age/sex	1 mmol/L	0.134019	0.024079	−0.109939
CHNS age, adjusted sex/BMI/glucose	1 year	0.007281	0.004854	−0.002426
ENSANUT age, adjusted sex/BMI/glucose	10 years	0.063944	0.024670	−0.039274
ENSANUT BMI, adjusted age/sex	5 kg/m²	0.056883	0.023732	−0.033151
ENSANUT BMI, adjusted age/sex/glucose	5 kg/m²	0.048797	0.025649	−0.023147



Figure 2: Relational evidence and scientific analyses across two cohorts. Panels (a–c, e–g) retain all six prespecified native-unit effects, five Agent constructions per condition and the separate statistical constructions. E0 uses basic/marginal evidence; E1 adds relations. Agent rows run A01–A05 from top to bottom, without pairing across conditions. Dots are three seed estimates, diamonds are constructor means and segments are seed ranges. Dashed reference estimates and gray bands show CHNS 95% community-cluster CR1-t intervals or ENSANUT cluster-resampling stability bands. HbA1c effects are percentage points per the stated increment. Panels (d, h) show 100 × (MAE_E1 − MAE_E0) / MAE_E0 for each endpoint. MAE averages seed absolute discrepancies within each constructor, then across five constructors; negative values indicate lower discrepancy. BMI + G denotes additional glucose adjustment. All 24 constructions, 72 final tables and 216 effects remain, including the CHNS BMI deterioration. These descriptive contrasts and bands do not define equivalence or a pooled score.

ative mean and two negative seeds. All ENSANUT E1 focal coefficients were positive, although its age means of 0.097769–0.102452 per decade exceeded 0.075132 and its BMI means remained below their references. Preserving a

direction and recovering its magnitude are distinct aspects of usefulness.

Every E1 author used at least one relationship as a numerical constraint. The complete uptake records include constraints, unfitted analytic checks and contextual uses, including the projection calculation in Section 3.2. Statistical references also achieved useful results: under E1, the prescribed constructions had lower seed-averaged discrepancies than every corresponding Agent for CHNS glucose and ENSANUT conditional BMI. Other comparisons favored Agent constructions. The evidence therefore supports scientific utility and the value of additional information, rather than a universal algorithmic ranking.

4.3 Q2: accessible summaries become useful joint data

The separate CHNS experiment gave three authors external literature (L) and three authors that literature plus 30 target summaries and their metadata (R). Each author delivered three tables and had one additional diagnostic-review opportunity. All six retained their models. There were 42 physical executions: six trials and 36 initial/final deliveries. Final bytes duplicated their matching initial outputs, leaving 24 distinct tables overall and 18 distinct delivery tables. The identical stages are displayed once.

R supported close reconstruction of all five highlighted scientific analyses (Table 3; Figure 3). These values were legitimate construction inputs; their agreement measures reconstruction rather than prediction of hidden coefficients. The L/R packet and authors differ from Q1, so those two information comparisons are kept separate.

Agreement extended beyond central values. R constructor-mean BMI SDs were 3.38965–3.40243 versus 3.40632, glucose SDs were 1.48817–1.58968 versus 1.54455, and HbA1c SDs were 0.95208–0.96554 versus 0.95357. A separate three-table statistical calibration supplement also closely reconstructed the regression summaries, with mean coefficients 0.055726, 0.412568 and 0.007679. That purpose-built family was developed with coding-Agent assistance and is not an Agent-removal ablation. Both approaches demonstrated the usefulness of constructing a joint table from accessible aggregate information.

4.4 Q3: a published model supports longitudinal analysis

The HRS recipient-stage example [1] tested a different data product. A fresh Agent received published model exports, a variable specification and the intended mixed-model analysis. It authored the recursive conditional generating family with parameter provenance. A separate implementation reconstructed the published law. Each generated 11,509 subjects at 0, 2, 4 and 6 years, producing 46,036 rows per table. No HRS individual records were used, and the source analysis coefficients were construction inputs.

All three Agent and three corrected-law tables supported the native R/lme4 model with 16 fixed effects, individual intercepts/slopes and piecewise HbA1c terms around 6.5%, with age centered at 60 and scaled per decade. The below-threshold HbA1c-by-age interaction averaged −0.044635 in Agent data, with range −0.051059 to −0.038247, against the supplied −0.046837. The reconstructed law averaged −0.040500. Both preserved its negative direction in all three draws (Figure 4).

The weaker above-threshold interaction was less stable: Agent estimates averaged 0.026485 against 0.012867, while corrected-law estimates averaged −0.011691. Overall Agent HbA1c mean/SD was 5.8725/0.9676 versus the original published 5.9/1.0. The high-HbA1c subgroup had mean/SD 7.1515/0.5630, compared with original 7.7/1.4 and published synthetic 7.2/0.6. Thus useful source-model behavior coexisted with inadequate upper-group spread.

A CRLF-sensitive parser initially overwrote seven coefficient/SE pairs in our comparator preparation. A separately reviewed source-bound repair produced the three corrected comparator tables without changing the frozen Agent outputs. The full record contains nine physical generations, six valid comparison tables and three invalid initial comparator attempts. This was our implementation defect, not a failure of the published method. The incident record and complete effects retain the correction and all outcomes.

4.5 Q4: open search leads to data and a precise evidence gap

The open-search request concerned `HbA1c ~ BMI + glucose + age + female` among Chinese adults aged 35–74 around 2009. A fresh acquisition Agent received no bibliographic seeds, desired signs or generator family. Ten recorded queries led to five seriously reviewed originals, with CHNS-derived numerical evidence excluded. The source review distinguished continuous conditional coefficients [9] from grouped contrasts [14], unclear-scale coefficients [13] and diabetes logistic effects [12]. Pani supplied complementary age evidence [10]. Population and adjustment differences remained visible.

A new scientific constructor used that packet to choose a sex-stratified Gaussian copula, bounded integer age and positive lognormal/shifted-lognormal margins. It converted Pearson and Spearman evidence under the selected marginal laws, used source summaries with explicit transport assumptions and retained less directly applicable conditional findings as context. No scientific parameter or target answer was supplied by the controller. One trial and three fixed deliveries succeeded, each with 3,000 complete rows. Each execution returned 11 approximate and 23 unavailable source comparisons; after feedback the Agent retained its model.

The resulting data supported useful secondary analyses without glucose adjustment. BMI coefficients were 0.04217–0.05176 versus 0.05495 in the reused CHNS reference, and age coefficients were 0.01613–0.01771 versus 0.01530. Primary age was also close at 0.00842–0.01084 versus 0.00811. Primary BMI and glucose were attenuated, and the primary female point direction differed from its uncertain reference. Figure 5 retains all seven nonintercept coefficients,

Table 3: Joint-statistic reconstruction with available summaries. Ranges span three constructor means over three fixed seeds, not confidence intervals. Regression units are HbA1c percentage points per kg/m², mmol/L and year, respectively.

Analysis	Supplied reference	External literature L	Literature + summaries R
BMI coefficient, age/sex adjusted	0.054946	0.013001–0.034287	0.052821–0.055789
Glucose coefficient, age/sex adjusted	0.402093	0.356064–0.487815	0.396941–0.413715
Age coefficient, sex/BMI/glucose adjusted	0.008107	0.003552–0.006910	0.007569–0.007947
Glucose–HbA1c Pearson correlation	0.660968	0.633764–0.813170	0.650561–0.666438
Glucose–HbA1c Spearman correlation	0.361457	0.501339–0.566477	0.357571–0.360983

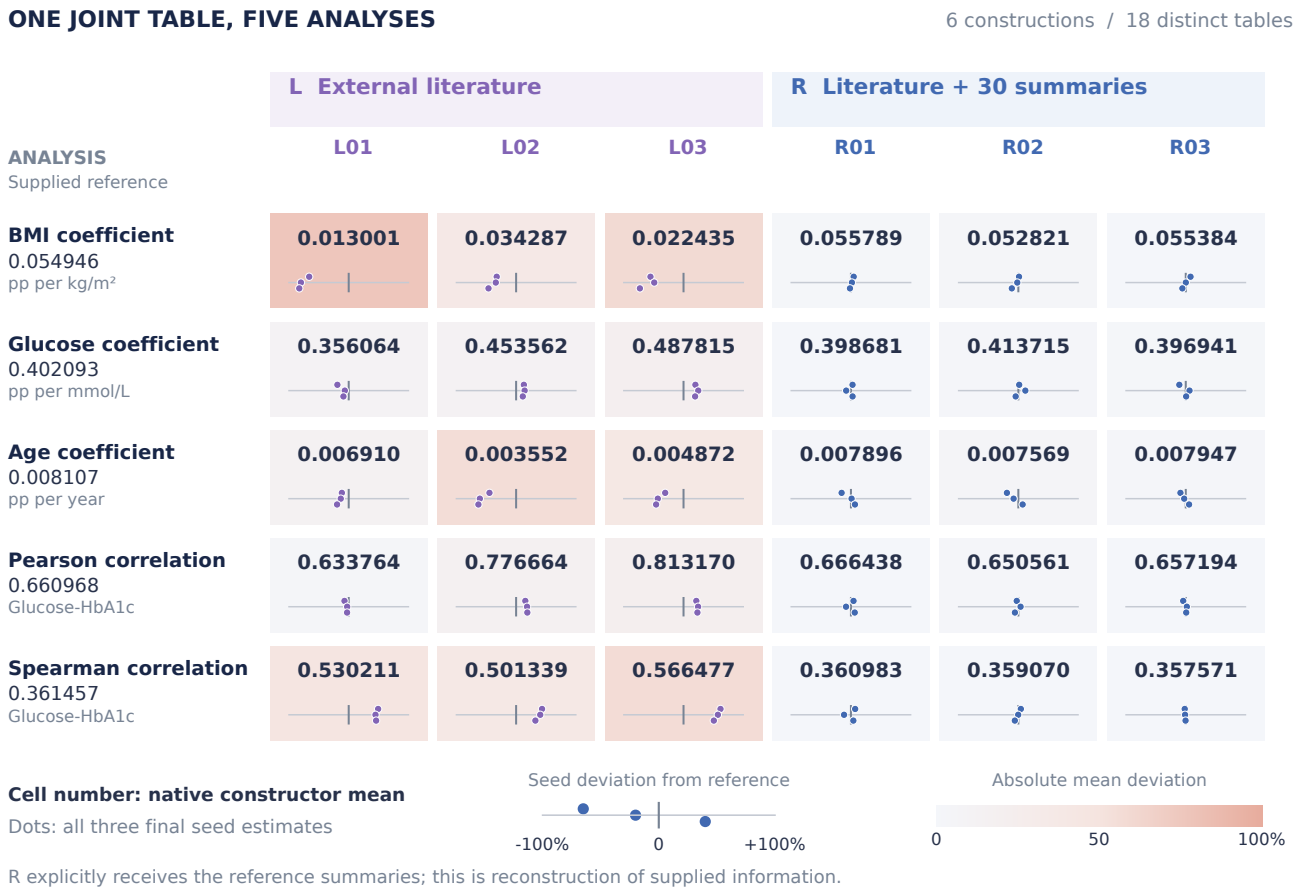


Figure 3: Joint reconstruction from supplied summaries. Columns represent six independent constructions and rows represent five analyses of the same final tables. L uses literature; R additionally receives 30 target summaries. Cells print native-unit three-seed means. Regression rows adjust BMI or glucose for age/sex, and age for sex/BMI/glucose; correlation rows concern glucose–HbA1c. Background intensity is 100 × |mean − reference| / |reference|. Three dots retain every seed on the common signed track 100 × (estimate − reference) / |reference|; its endpoints are −100% and +100% and its center is zero. Vertical offsets distinguish seeds only. All 30 means and 90 seed statistics from 18 distinct final tables are shown. Normalization is a display convention; reference uncertainty is retained in the numerical supplement rather than plotted here. R reconstructs explicitly supplied information. Identical initial/final stages are displayed once, without implying revision gain.

their uncertainty labels and every fixed seed. Distributional differences also remained, including excess female proportion and a higher HbA1c 90th percentile; the full diagnostic table is preserved.

The sensitivity was identified before generation or target comparison. The delivered model's dependence of 0.25 was an explicit assumption. Both alternative covariance structures were valid, and independent calculation reproduced the sign change. This identifies the particular missing relationship that affects the requested conditional analysis even when other fitted relationships stay unchanged. It does not establish that one assumption explains every observed discrepancy.

Real-sample variation provides additional context. A separately frozen analysis partitioned the existing CHNS communities into disjoint halves 100 times and sampled 3,000 people per side. For statistic T, it compared $|T(R1) - T(R2)|$ with the symmetric generated discrepancy $(|T(G) - T(R1)| + |T(G) - T(R2)|)/2$, retaining all 36 metrics. For secondary BMI, the real-real median was 0.007239 versus generated symmetric medi-

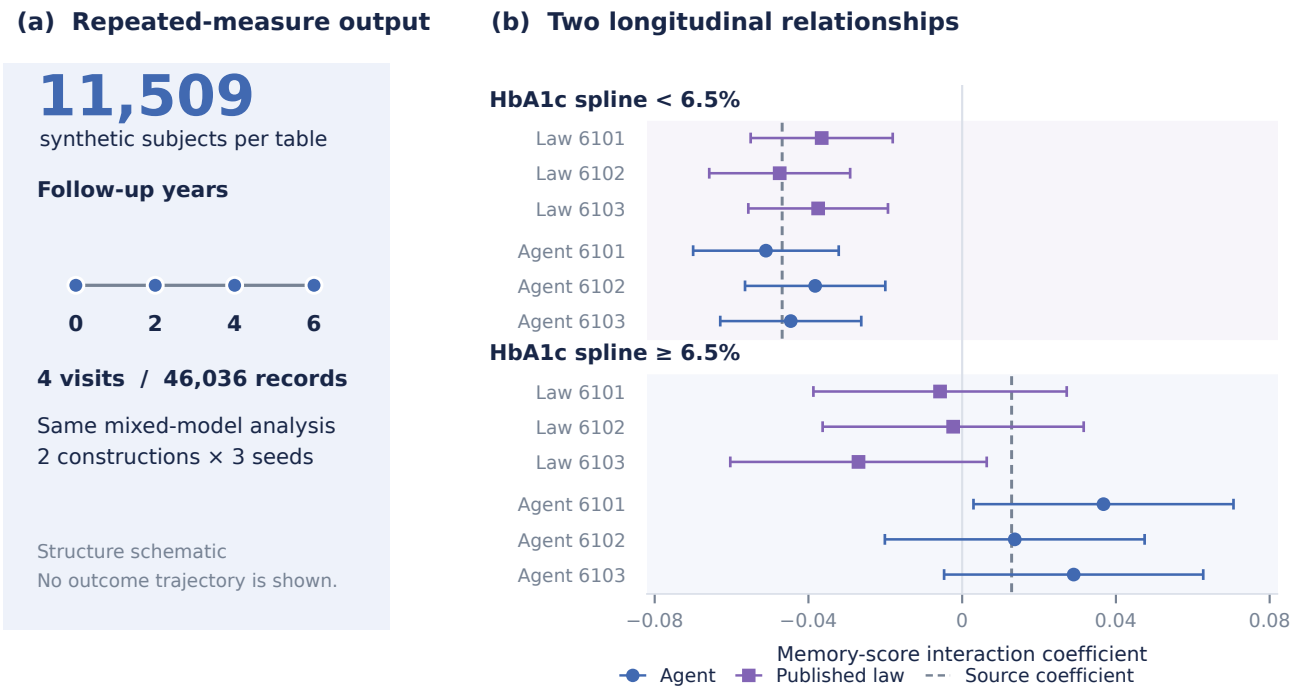


Figure 4: Longitudinal data and analysis preservation. Panel (a) shows the actual four-visit structure: 11,509 virtual subjects and 46,036 records per table. It illustrates scheduling, not outcome trajectories. Panel (b) retains both HbA1c-spline-by-age interactions from the same full mixed model, with three seeds per construction. Blue circles and violet squares show stored estimates and conditional 95% Wald intervals; dashed lines show source coefficients and the solid line marks zero interaction. Coefficients are memory-score change per decade per HbA1c percentage-point spline increment. Law denotes the corrected published-law reconstruction. Both methods receive published model artifacts; this is source-rich reconstruction, not independent clinical confirmation. All 16 effects, source uncertainty, upper-HbA1c spread discrepancy and invalid initial comparator attempts remain in the supporting record.

ans of 0.004383–0.012727 across the three tables; for secondary age, the corresponding values were 0.001705 and 0.001055–0.002387. Primary glucose and female discrepancies were substantially larger than their resplit context. These nonexchangeable finite-sample comparisons help interpret analysis-specific usefulness without defining a retrospective equivalence threshold. The complete table and seven-term supplement retain every result.

5 Discussion

The experiments support a concrete role for a scientific Agent: construct data that researchers can analyze by selecting and numerically reconciling partial biomedical evidence. The important output is a joint data product, not a predicted answer to one regression. Cross-sectional and longitudinal examples show that this product can preserve useful analytical relationships under several information conditions. The source-to-model traces explain what the Agent contributed between a paper's result and a generated table.

The strongest methodological lesson is that scientific use depends on the relationship being asked about. External relations improved five of six main-study contrasts, and provided summaries supported close joint reconstruction. Yet a model matching several margins or pairwise relations can still imply a different conditional effect. The copula sensitivity makes this gap explicit and suggests a focused next action: acquire joint or subgroup evidence for the dependence that controls the requested analysis. This diagnostic use of an executable model can guide research even when the available evidence does not identify a unique target distribution.

The framework's statistical operations build on established mathematics and related LLM construction methods [3–6]. Its applied contribution connects a biomedical request to qualified evidence, scientific model authorship, data generation and analysis-centered evaluation. The current experiments demonstrate particular uses; they do not establish uniform superiority over task-specific statistics. G-Sim and LLMSynthor also delegate consequential model or data-construction choices to LLMs, so the presence of an Agent or a feedback loop alone is not a field-priority claim.

Three limits determine the scope of the evidence. First, CHNS and ENSANUT are exposed developmental references with partly shared sources; later CHNS work reuses that reference. Summary-assisted R and the HRS model intentionally receive the relevant numerical answers. These information settings remain distinct from independent transport validation. Second, generated seed ranges and conditional model intervals omit source-estimation, extraction, structural and transport uncertainty. The real-real analysis describes variation within one selected sample, not between populations. Third, actual workflow integration remains host-dispatched and coordinator-managed. Autonomous execution for arbitrary requests, preparation-time savings and broad-domain

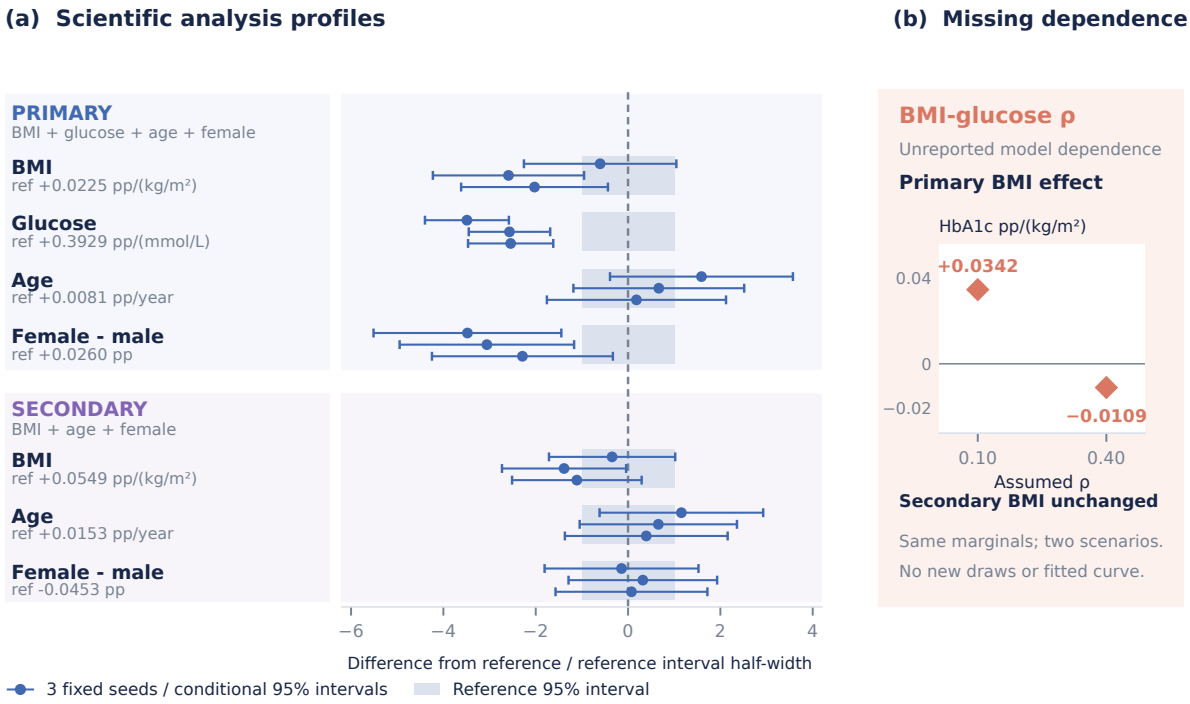


Figure 5: Scientific analysis profiles and a missing joint relationship. Panel (a) shows all seven effects from three 3,000-person tables, with native reference values and units. For display, each estimate and interval bound becomes (value − reference estimate) / ((reference upper bound − reference lower bound) / 2). This coordinate is not a test or acceptance criterion; dashed zero denotes the sample reference. Seeds 7401–7403 run top to bottom with conditional 95% HC3-t intervals. Gray bands transform the CHNS community-cluster CR1-t bounds separately. Generated intervals omit source, structural and transport uncertainty. Panel (b) shows two pre-generation analytical scenarios: latent BMI–glucose correlation 0.10 or 0.40 yields primary BMI slopes +0.03422 or −0.01095 pp/(kg/m²). Secondary BMI, margins and other bivariate laws remain fixed. The delivered model uses 0.25. These are assumption-dependent calculations, not additional samples or uncertainty limits.

scalability have not been measured.

Feedback is operational but its scientific benefit is unresolved. Seven main-study authors reviewed exploratory outputs and retained their laws; thirteen finalized without exploration. All six L/R authors and the latest constructor retained their models after review. Earlier live authors likewise retained their outputs, although a coordinator-detected divergent-moment defect was repaired by the author before freeze. Those records establish meaningful inspection and valid retention, not a general revision-gain result. Earlier enrichment, direct-row construction, failed transport criteria, tail defects and revision trade-offs remain in the historical supplement; missing-cell completion is unvalidated.

The next discriminating extension is a prospectively frozen request-to-data workflow across independently defined tasks and unexposed references, with actual effort and uncertainty recorded. It should test specific claims about additional evidence or workflow reuse. The present paper already establishes a narrower useful capability: published evidence can be made executable as virtual biomedical data, and an Agent can expose the assumptions that determine which scientific analyses those data support.

6 Supporting results, code and reproducibility

The public repository provides original code, protocols, aggregate results and independent checks. The complete v0.9 extended record preserves all six original numerical tables and full methods/results. Its Table 2 supplies every main endpoint's adjustment set, units and reference interval; Tables 3–4 retain the main and summary-assisted values; Table 5 retains all latest seven effects; Table 6 retains the real-real comparison values. No result was discarded by this narrative condensation.

The main study, summary-assisted study, statistical supplement, HRS example, open search and latest construction preserve separate frozen identities. Earlier live A/B results, including the conditioned BMI mismatch and source-replay unavailability, remain in the extended record. Numerical and visual reproduction is aggregate-only and does not constitute a new scientific experiment.

Individual real/generated tables, third-party originals and credentials remain local; clinical reruns require lawful access and the appropriate environment. Historical model identifiers and token costs are incompletely metered. Author/affiliation/funding/conflict/ethics/access/AI declarations and final submission formatting remain to be completed. No manuscript has been submitted.

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